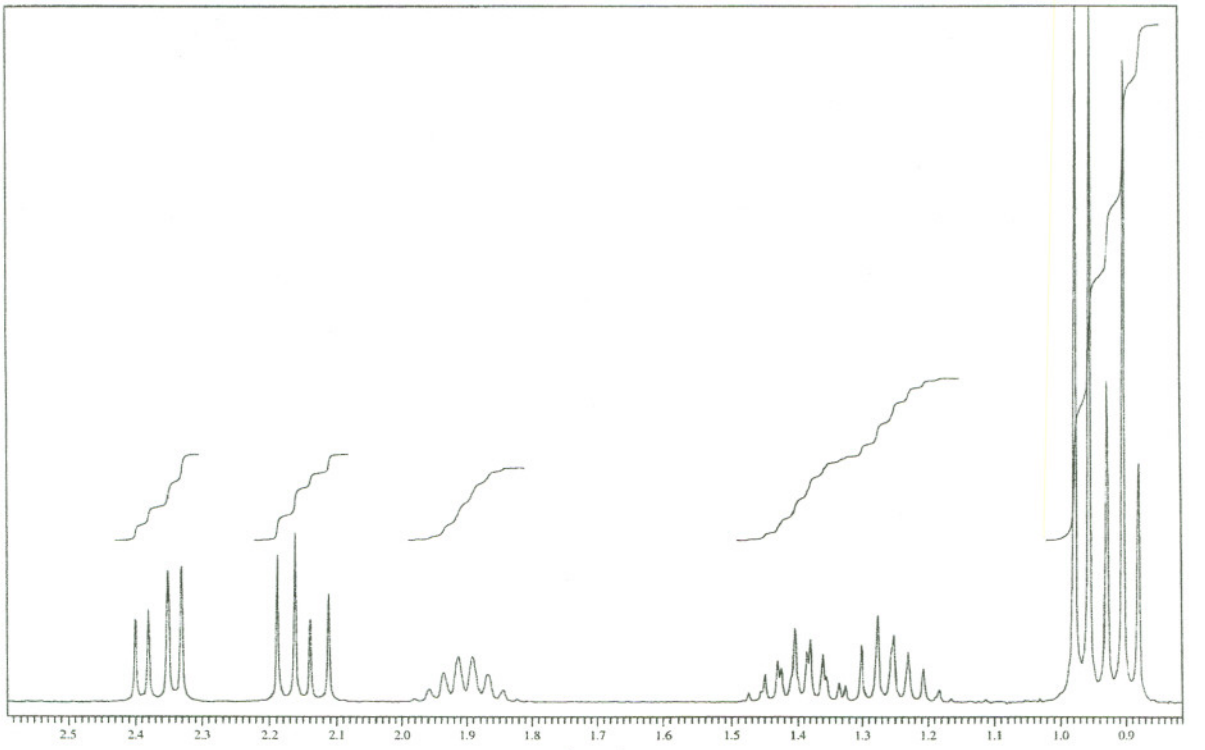
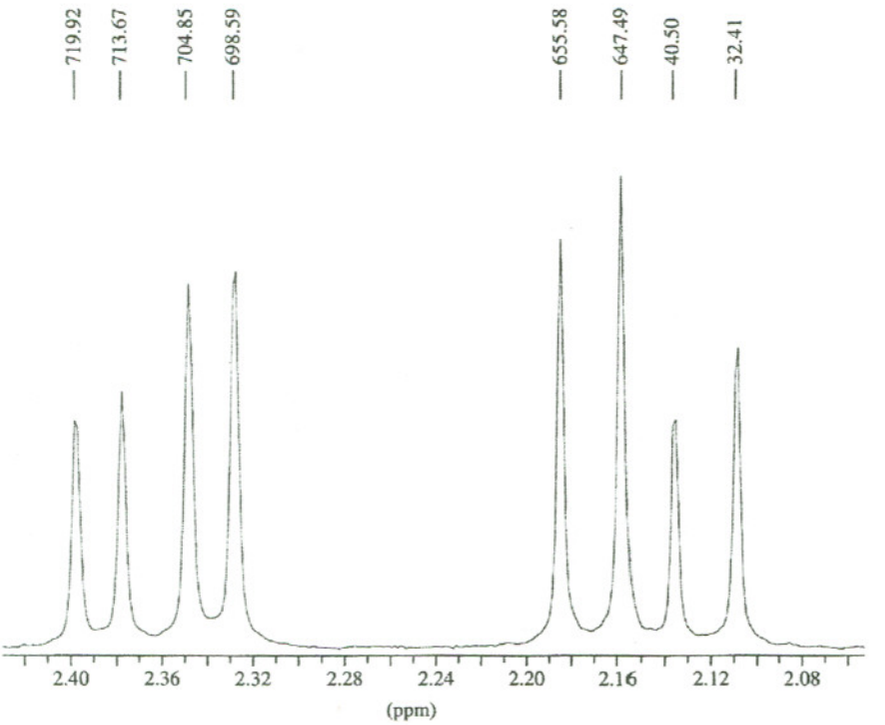
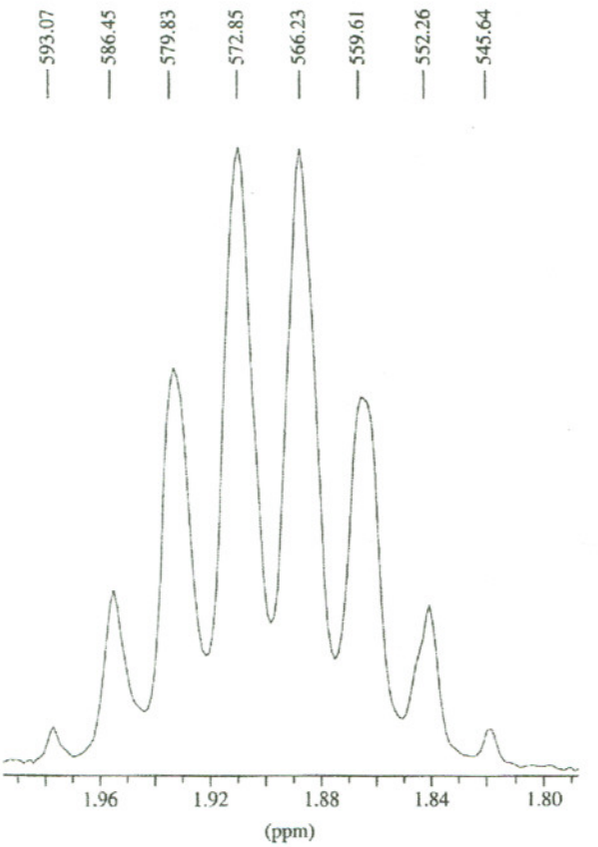
ORGANIC CUME
January 24, 2003

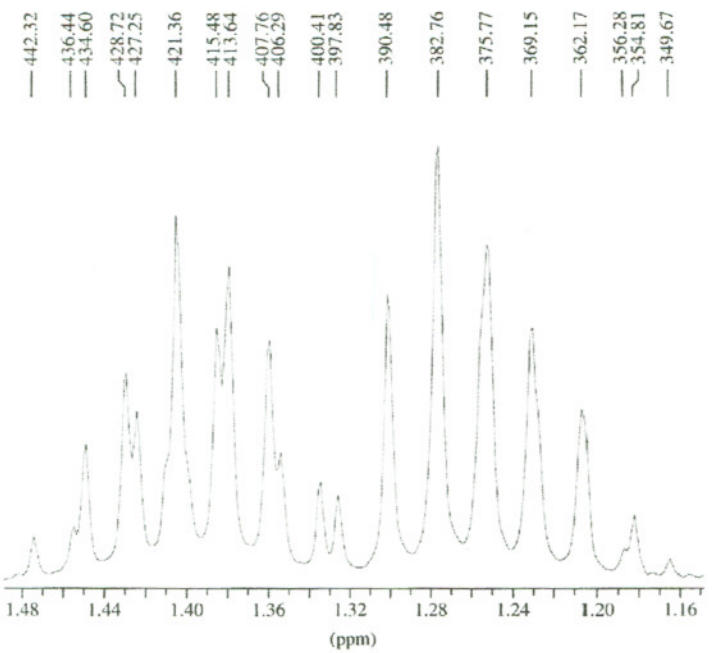
1) Determine the structure for a compound with the molecular formula $C_6H_{12}O_2$. The IR spectrum shows a strong and broad band from 3400-2400 cm^{-1}, and also at 1710 cm^{-1}. The ^1H NMR spectrum is provided, but the peak at 12.0 ppm is not shown. The HETCOR is also given. Fully interpret the proton NMR spectrum and assign all of the protons and carbons for this compound.
(helpful info: Hydrogen Deficiency Index = [2C + N + 2 – X – H]/2, where C is the number of carbon atoms, N = number of nitrogen atoms, X = halogens, etc.).



(ppm)

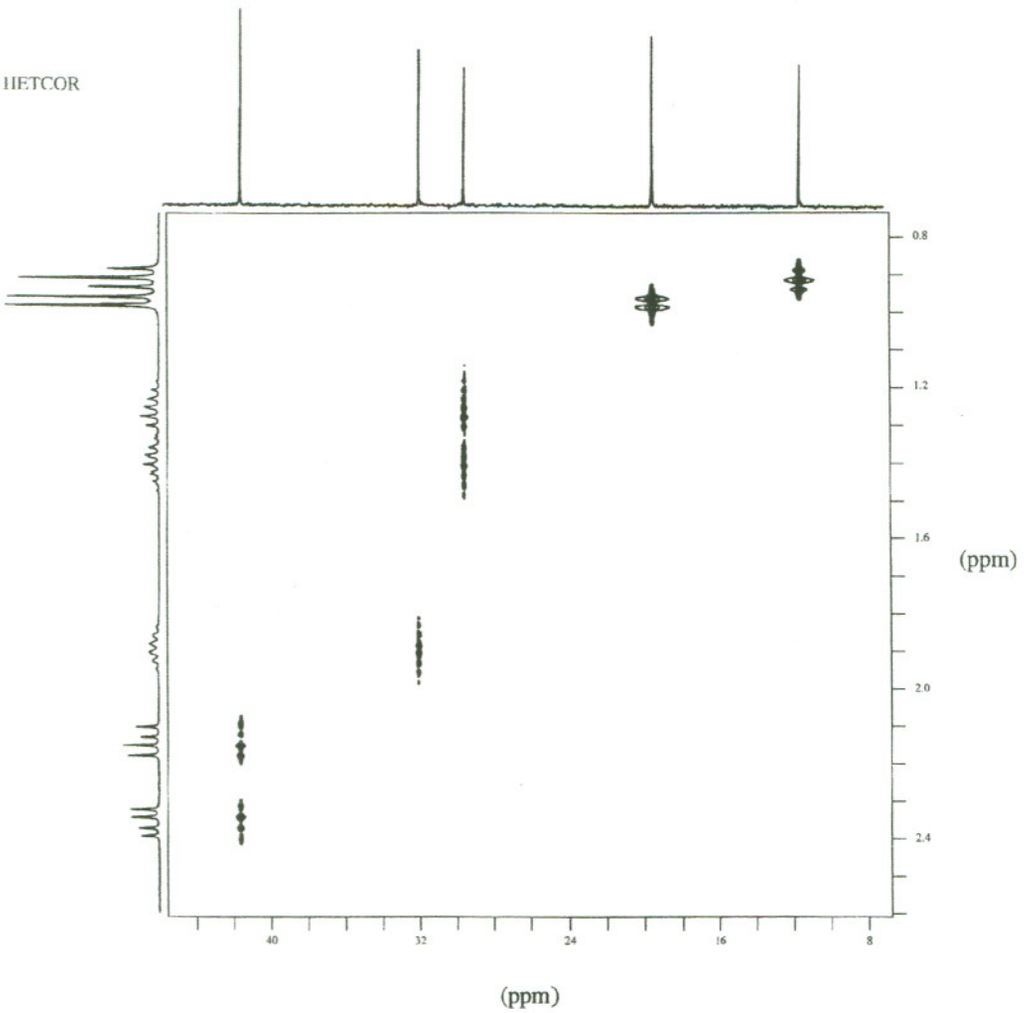

719.92
713.67
704.85
698.59
655.58
647.49
40.50
32.41
2.40
2.36
2.32
2.28
2.24
2.20
2.16
2.12
2.08
(ppm)


593.07
586.45
579.83
572.85
566.23
559.61
552.26
545.64
1.96
1.92
1.88
1.84
1.80
(ppm)


442.32
436.44
434.60
428.72
427.25
421.36
415.48
413.64
407.76
406.29
400.41
397.83
390.48
382.76
375.77
369.15
362.17
356.28
354.81
349.67
1.48
1.44
1.40
1.36
1.32
1.28
1.24
1.20
1.16
(ppm)

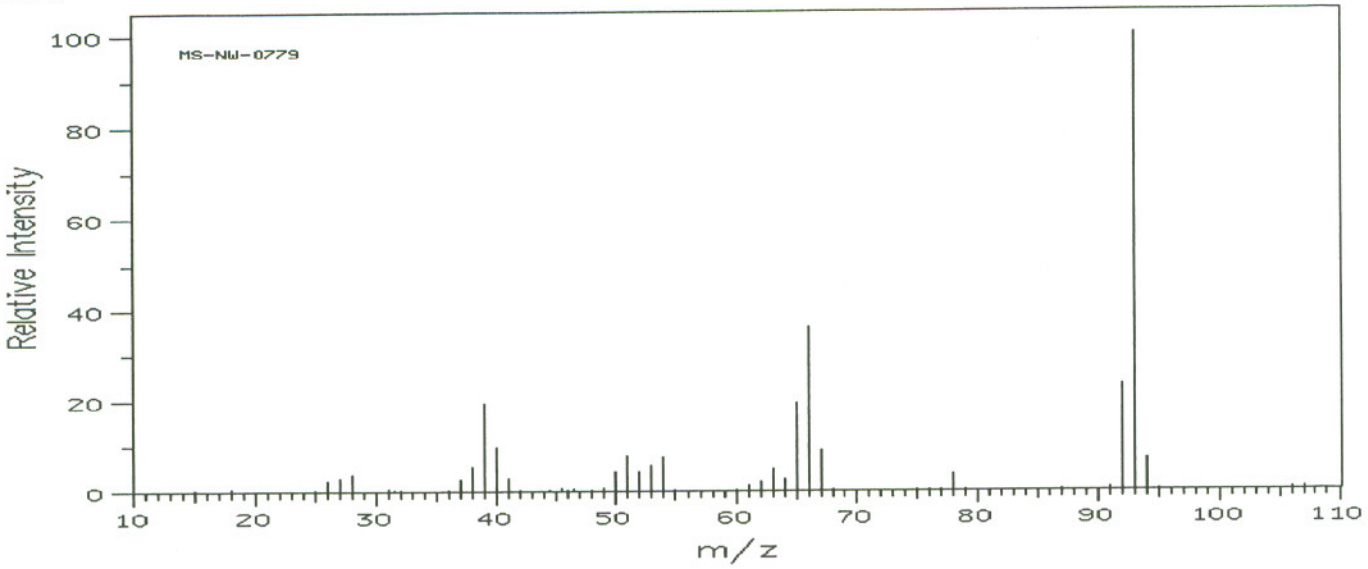
294.15
287.53
279.81
272.45
264.73
1.00
0.96
0.92
0.88
(ppm)

HETCOR

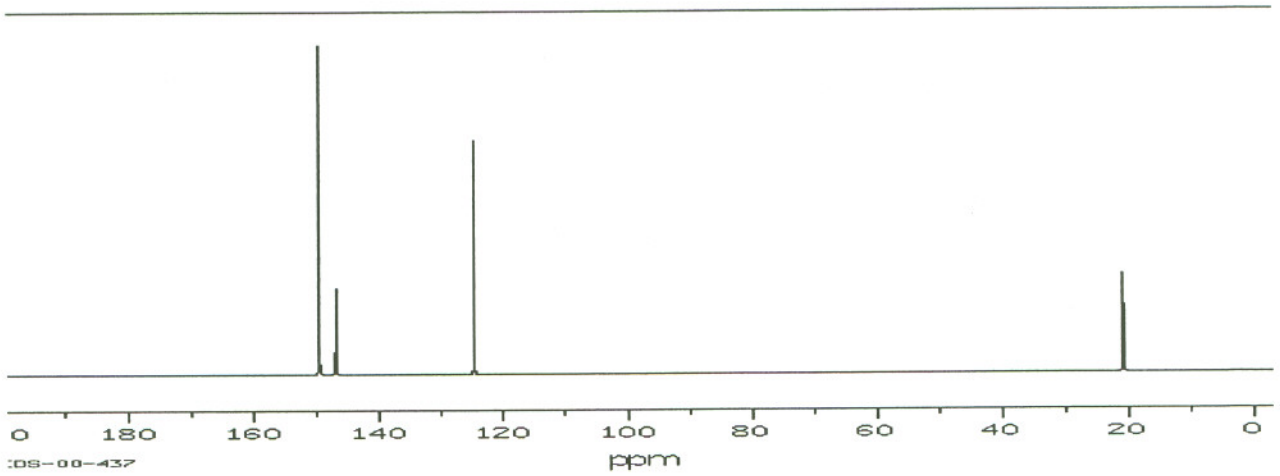


Problem 3:

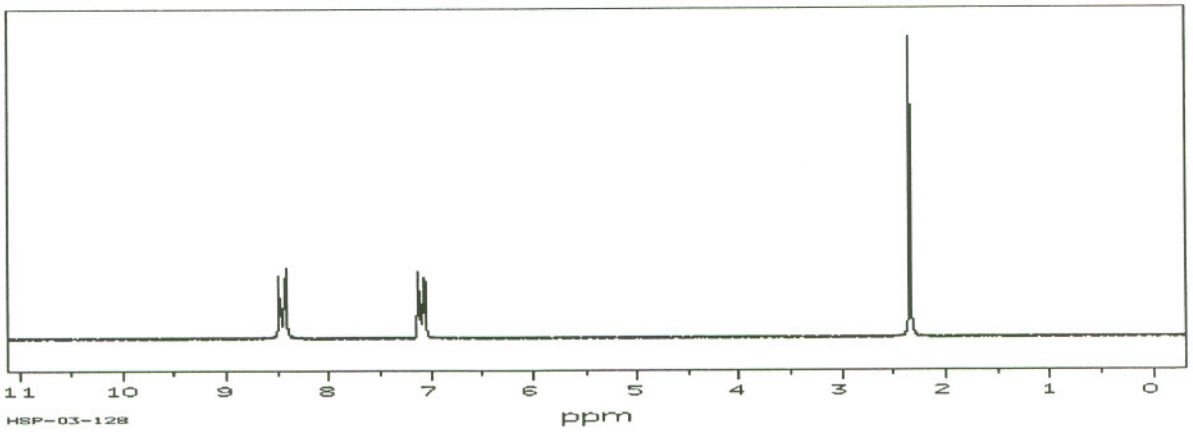
MS:



^{13}C NMR:
25.16
MHz

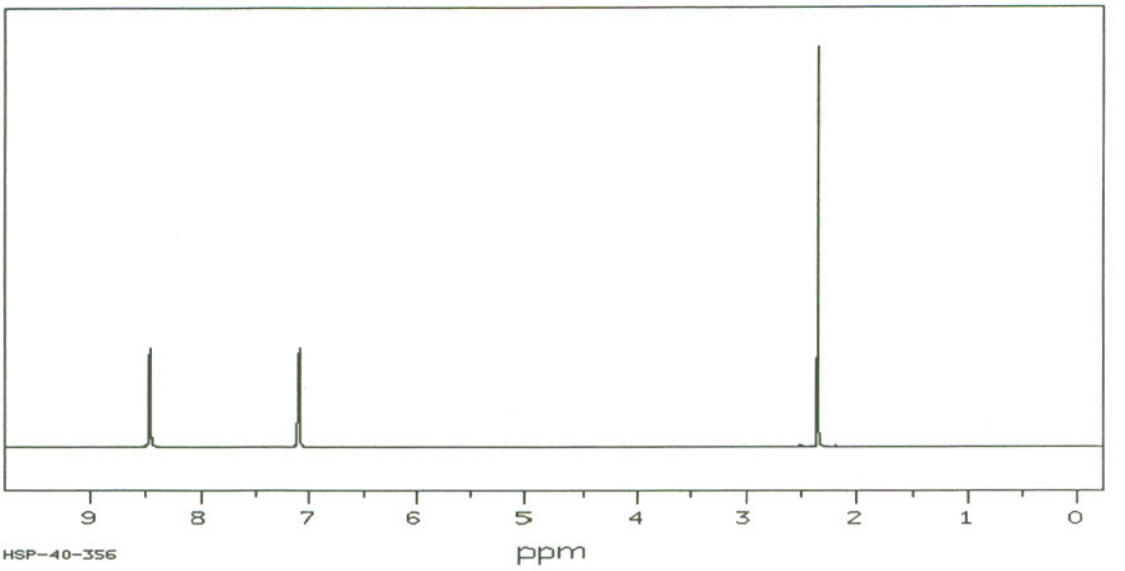


^{1}H NMR:
90 MHz

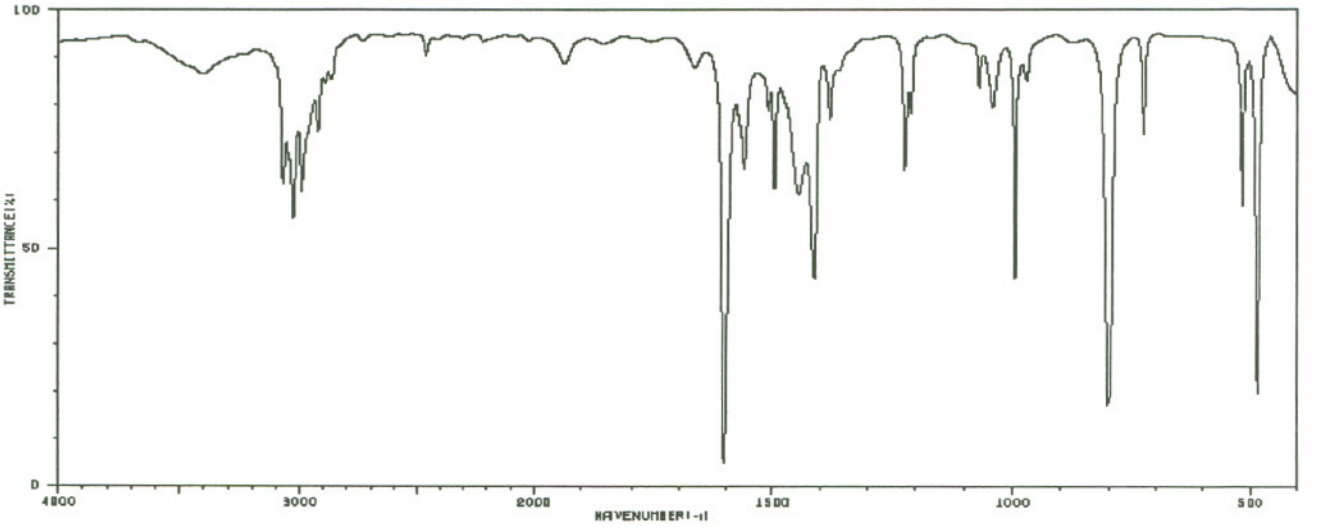


Problem 3 continued:

^{1}H NMR:
400 MHz
Integration of absorptions (downfield to upfield, 2:2:3)



IR:
liquid film



UV spectrum: absorptions observed above 200nm